

DUONG QUOC VINH

AI Engineering Intern | LLM & RAG Focus

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Portfolio: dquocvinh.github.io/Portfolio

JOB OBJECTIVE: Seeking a Full-time AI Engineering Internship in Information Technology.

PROFESSIONAL SUMMARY

- Final-year Computer Science student (AI Major) at Ton Duc Thang University targeting an AI Engineering Internship.
- Experienced in engineering end-to-end LLM/RAG applications, fine-tuning transformers (BERT), multimodal AI, and latency optimization.
- Proficient in PyTorch, LangChain, FAISS/Pinecone, FastAPI, and React/TypeScript to deliver scalable, production-ready AI solutions.

RELEVANT PROJECTS

A Machine Learning Approach to Psychological Disorder Screening & Virtual Assistant for Mental Health Support (2025-2026)

[\[GitHub\]](#)

Tech Stack: Python, PyTorch, BERT, LangChain, Pinecone, ChromaDB, FastAPI, React, Next.js

- Built a 5-stage clinical AI system integrating PHQ-9/GAD-7 surveys with a fine-tuned BERT model (103K+ records, 83.6%–88.8% accuracy) for 7-class emotion screening.
- Developed a CBT-grounded RAG virtual assistant (LangChain, Pinecone) featuring a MIND-SAFE crisis override gate to automatically reroute high-risk queries to emergency hotlines.
- Deployed async RESTful inference APIs via FastAPI integrated with a React/Vite dashboard for real-time risk assessment and interactive user visualization.

AI Portfolio Assistant & RAG System (2026) [\[GitHub\]](#)

Tech Stack: React, TypeScript, FastAPI, LangChain, FAISS, Gemini API, Vite, GitHub Pages, Render

- Developed a bilingual (EN/VI) portfolio RAG chatbot combining FastAPI, local FAISS vector store, and Gemini 3.1 Flash Lite LLM with a 3-stage chunking & embedding pipeline.
- Cut pipeline latency by ~90% (42s down to 3–5s) by migrating to Google Embedding API and local FAISS, eliminating cold-start overhead and network round-trips.
- Built an interactive React/TypeScript UI with typewriter streaming, custom Markdown rendering, dynamic URL links, and automated CI/CD deployment via GitHub Pages and Render.

Vi-VQA Animal – Multimodal Visual Question Answering System (2026) [\[GitHub\]](#)

Tech Stack: Python, PyTorch, ResNet50, BiLSTM, Transformer Decoder, Co-Attention Fusion, BLIP

- Engineered a multimodal Seq2Seq VQA framework combining ResNet50 vision backbone, BiLSTM question encoder, and Co-Attention Fusion module across 32,400 samples.
- Integrated a Transformer answer decoder achieving benchmark performance (0.5778 Exact Match, 0.9062 BERTScore), significantly outperforming baseline models and zero-shot BLIP.

EDUCATION

Bachelor of Science in Computer Science (AI), Ton Duc Thang University (*Expected 2027*)

TECHNICAL SKILLS

- **AI & LLM Core:** Large Language Models (LLMs), RAG, GraphRAG, Multimodal VQA, Vector Search, Latency Optimization.
- **Frameworks & Tools:** PyTorch, HuggingFace Transformers, LangChain, LlamaIndex, FAISS, Gemini API, Scikit-Learn.

- **Databases & Web:** FAISS, Pinecone, ChromaDB, PostgreSQL, Python, TypeScript, FastAPI, React.js, Next.js.
- **MLOps & DevOps:** Git/GitHub, GitHub Pages, Render, Docker, Linux, Kaggle Cloud GPU (Tesla T4), Model Evaluation Metrics.

CERTIFICATIONS & LANGUAGES

- **English Proficiency:** APTIS ESOL B2 (Equivalent to IELTS 5.5) – issued by British Council
- **Google Workspace with Gemini** – Authorized by Google & Offered through Coursera (2026) | [Credential Verification](#)
- **Google AI Certification** – Authorized by Google & Offered through Coursera (2026) | [Credential Verification](#)
- **Vibe Coding for Developers** – Offered through Coursera (2026) | [Credential Verification](#)